Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*9569837209*

**PHYSICS** **5054/41**

Paper 4 Alternative to Practical **May/June 2022**

**1 hour**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages. Any blank pages are indicated.


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**1** A student investigates how the temperature of a volume of water changes as hotter water is added to it.

160 $cm^3$ of water at 20 °C is poured into a glass measuring cylinder.

Fig. 1.1 shows the measuring cylinder.

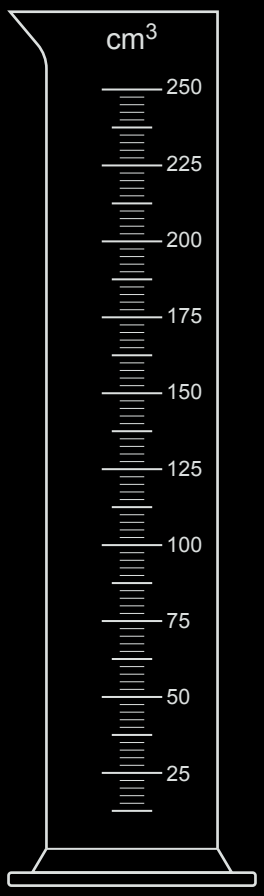



**Fig. 1.1**

**(a)** The measuring cylinder contains 160 $cm^3$ of water.

On Fig. 1.1, draw the level of the water, showing the meniscus. [2]

**(b)** The student pours 100 $cm^3$ of water from the measuring cylinder into a large beaker.

He then:

- uses a kettle to provide a constant supply of hot water
- adds 50 $cm^3$ of hot water at 70 °C to the beaker
- stirs the mixture
- records the new temperature of the water in the beaker
- continues to add 50 $cm^3$ of hot water at a time, recording the new temperature for each addition, until a total of 300 $cm^3$ of hot water has been added.

**(i)** The student ensures that the temperature of the water added each time is 70 °C.

Suggest how this is done.

.....

.....

.....

..... [2]

**(ii)** The student repeats the experiment and obtains three sets of results, as shown in Table 1.1.

**Table 1.1**

| total volume of hot water added $V$ / $cm^3$ | temperature $\theta$ / °C | | | average temperature $\theta_{av}$ / °C |
|---|---|---|---|---|
| 0 | 20 | 20 | 20 | 20 |
| 50 | 33 | 34 | 33 | 33 |
| 100 | 42 | 40 | 40 | 41 |
| 150 | 47 | 49 | 46 | |
| 200 | 51 | 52 | 51 | 51 |
| 250 | 53 | 54 | 53 | 53 |
| 300 | 54 | 55 | 54 | 54 |

Calculate the average temperature $\theta_{av}$ when total volume of hot water added $V$ = 150 $cm^3$.

Record, in Table 1.1, your answer to an appropriate number of significant figures.

[2]

**(iii)** On Fig. 1.2, plot a graph of $\theta_{av}$ / °C on the *y*-axis against $V$ / $cm^3$ on the *x*-axis. Start both axes from the origin (0,0).

Draw a smooth curved line of best fit. [4]

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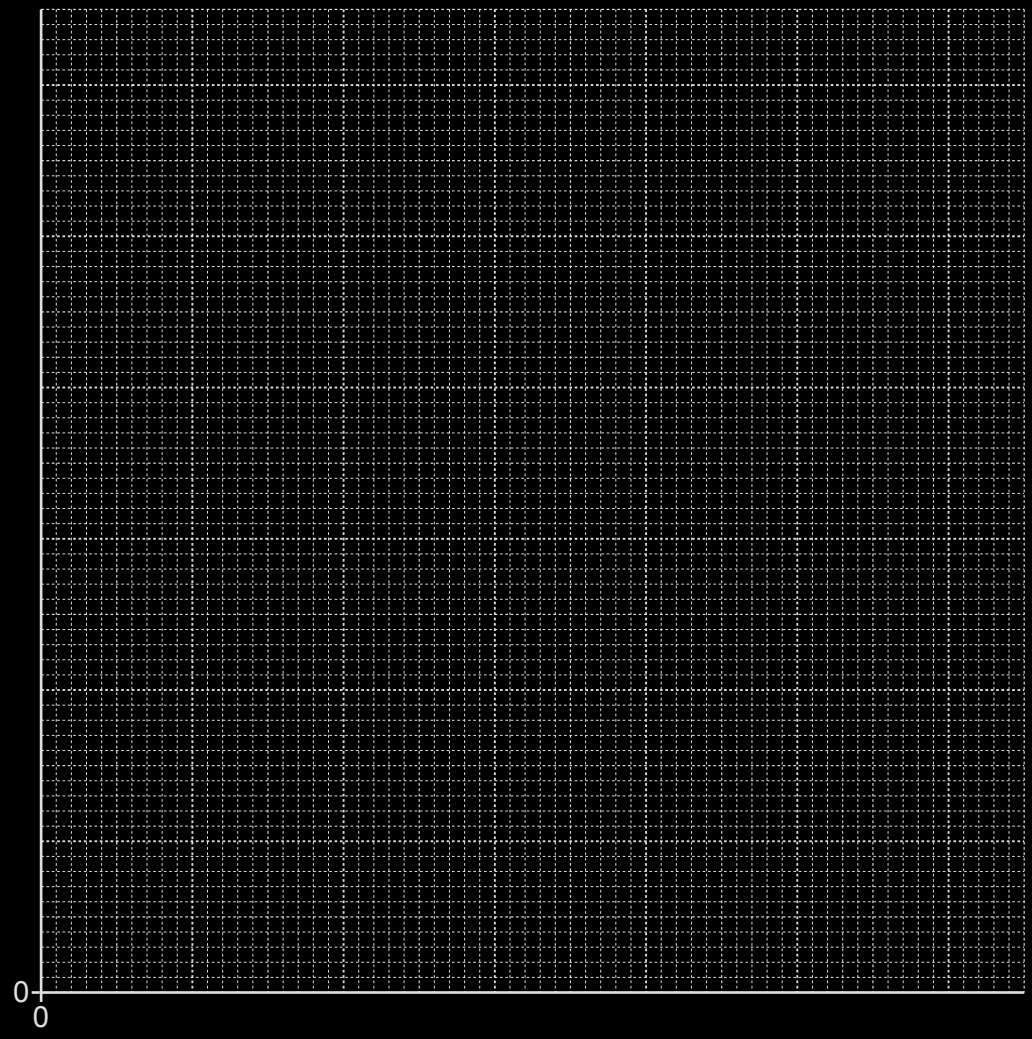



**Fig. 1.2**

**(iv)** Your graph shows that $\theta_{av}$ is **not** directly proportional to $V$.

Describe how your graph shows this and suggest why $\theta_{av}$ is not directly proportional to $V$.

.....

.....

.....

..... [2]

**(c)** The student repeats the experiment with a layer of insulation around the beaker.

On your graph in Fig. 1.2, sketch a line to show the results with a layer of insulation around the beaker. Label this line with the letter A. [2]

[Total: 14]

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**2** A student investigates the resistance of two resistors.

Fig. 2.1 shows the apparatus used to find the current $I$ in resistor X and the potential difference $V$ across it.

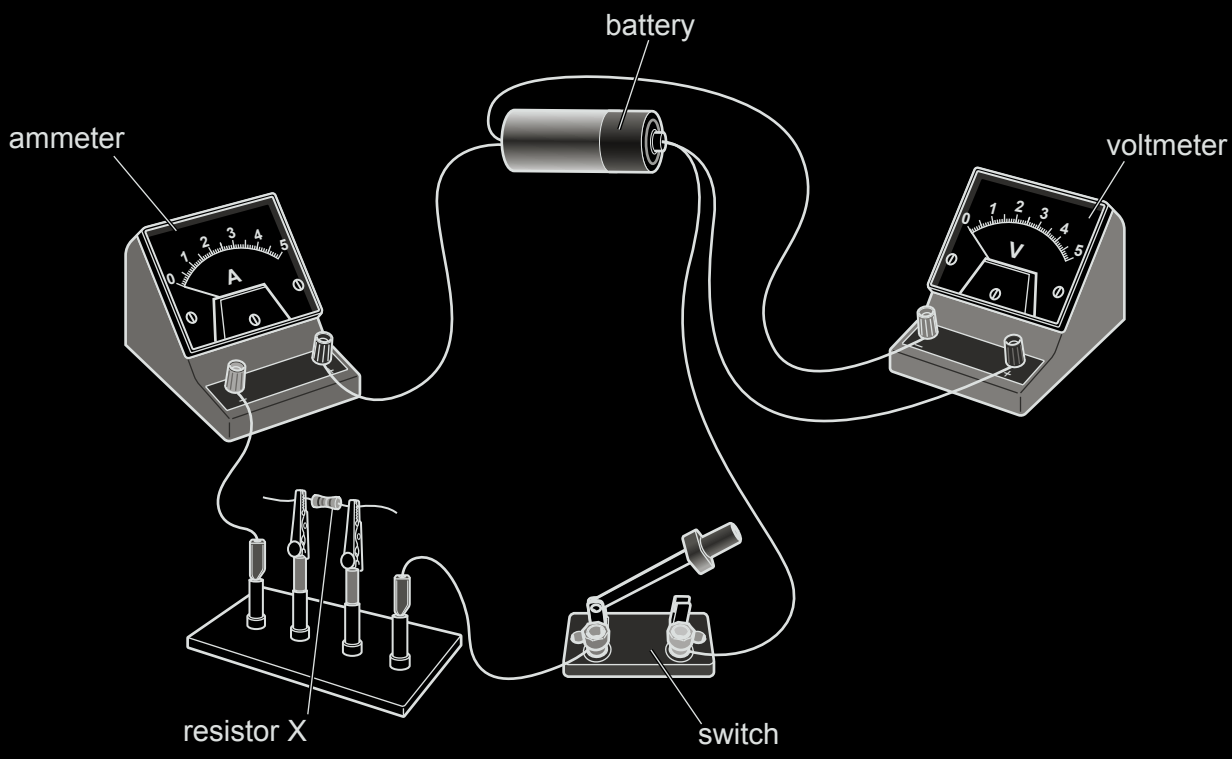



**Fig. 2.1**

**(a)** Draw a circuit diagram for the arrangement shown in Fig. 2.1.

[2]

**(b)** The student records the current $I$ and the potential difference $V$ across resistor X when the switch is closed. He calculates the resistance $R_X$ of resistor X.

He replaces resistor X with resistor Y and repeats the experiment.

The values recorded for both resistors are shown in Table 2.1.

**Table 2.1**

| resistor | $I$/A | $V$/V | resistance /Ω |
|---|---|---|---|
| X | 0.10 | 1.0 | 10 |
| Y | 0.07 | 1.0 | |

**(i)** Calculate the resistance $R_Y$ of resistor Y and complete Table 2.1.

Use the equation:

$$\text{resistance} = \frac{V}{I}$$

Give your answer to the nearest whole number.

[2]

**(ii)** The student combines resistors X and Y in a parallel arrangement. He uses this combination in the circuit in Fig. 2.1 in place of the single resistor Y.

Fig. 2.2 shows the ammeter reading for the parallel arrangement.

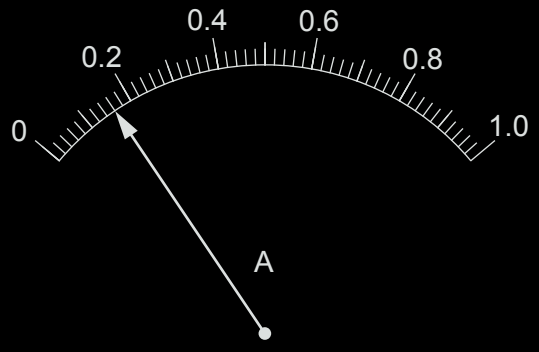



**Fig. 2.2**

Record the reading of $I$ shown in Fig. 2.2.

$I$ = ..... A [1]

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**(iii)** The potential difference stays at 1.0 V.

Using the value of $I$ in **(b)(ii)** and the equation in **(b)(i)**, calculate the resistance $R_C$ of the parallel combination.

$R_C$ = ..... Ω [1]

**(c)** Theory suggests that the resistance $R_C$ of the two resistors X and Y arranged in parallel is given by:

$$R_C = \frac{R_X R_Y}{R_X + R_Y}$$

State, giving a reason, whether your value for $R_C$ in (**b)(iii)** agrees with this suggestion.

.....

..... [1]

[Total: 7]

**3** A student directs two parallel rays of light, ray 1 and ray 2, towards the side of a transparent plastic block, as shown in Fig. 3.1. The block rests on a sheet of paper.

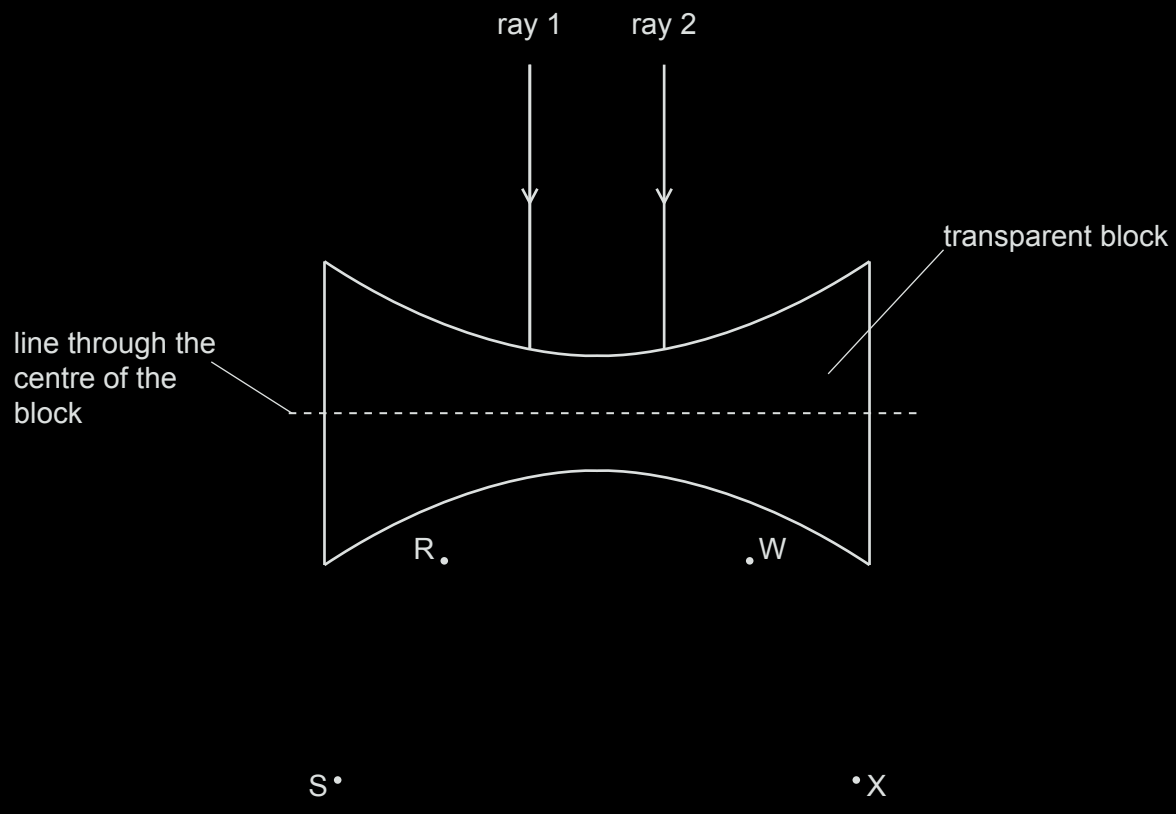



**Fig. 3.1**

**(a)** The student marks two points R and S on the path of ray 1 after it has passed through the block.

**(i)** On Fig. 3.1, draw a line joining S and R and continue the line to meet the block. [1]

**(ii)** On Fig. 3.1, draw a line to complete the path of ray 1 through the block. [1]

**(b)** The student marks two points W and X on the path of ray 2 after it has passed through the block.

On Fig. 3.1, complete the path of ray 2. [1]

**(c)** **(i)** Extend lines SR and XW back until they meet at a point. Label this point F. [1]

**(ii)** Measure the perpendicular distance from F to the centre of the block.

distance = ..... cm [1]

[Total: 5]

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**4** A student determines the diameter of a thin wire.

The apparatus available is:

- 2.0 m length of thin wire
- wire cutters
- ruler

**(a)** Suggest how the student determines an accurate value for the diameter of the wire using **only** the apparatus listed.

Draw a diagram to illustrate your method.

.....

.....

.....

..... [3]

**(b)** A second student determines the diameter using a different piece of apparatus. His value for the diameter is more accurate.

State the name of the different piece of apparatus he uses.

..... [1]

[Total: 4]

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